

Inequality of Opportunity in Brazil: Chances of Social Mobility for Individuals with High Levels of Education

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Introduction, Object, and Hypothesis of the Research

This work constitutes part of the thesis I am developing, dedicated to examining the inequality of opportunity in Brazil, focusing on the chances of individuals with high levels of education entering or remaining in the middle class. In the context of the expansion and restructuring of the Brazilian educational system over the last 40 years, there have been significant advances, such as the universalization of primary education, the restructuring of secondary and technical education, as well as the expansion and greater fluidity of access to higher education. This expansion and restructuring of the Brazilian educational system also brought advances in the general schooling of the population, with particular emphasis on the significant increase in individuals with completed secondary and higher education. This data is obtained not only in general schooling, but also when we observe the average schooling of each socio-occupational category in Brazil.

These related factors raise several questions about the sociological analysis of the relationship between academic training and the labor market. In other words, the academic training obtained by individuals guarantees the best chances of social advancement for those with a high level of education. The possibilities for conducting such an analysis vary from differentiating between income and socio-occupational category, intra- or intergenerational social mobility, as well as choosing a hierarchical or topological analysis.

In this sense, the general objective of this work is to analyze the impact of education on the chances of intergenerational social mobility for the middle class, based on a topological analysis of the occupational categories organized in the Brazilian social structure.

I use the EGP category "Professionals and Administrators" as a parameter to characterize the middle class. In the Brazilian context, this characterization is established by high school education, income, and self-identification with this sector of society (Salata, 2015). Thus, the objective is to analyze whether the chances of individuals with a high level of education entering or remaining in the middle class vary over time. Based on an analysis of the impact of education on intergenerational social mobility, I work with

two hypotheses: With the expansion of the educational system in Brazil, educational credentials are still important assets for overcoming the barrier of social origins (social mobility); however, their relative value has been weakening over time, even for highly educated individuals. These processes are related to previous findings on status barriers (Salata, Sin Yi, 2022; Salata, Sin Yi, 2025) and to the credentialist theory of diploma inflation (Collins, 1979), specifically, to the institutional theory of Lester Thurow (1975) and Hirsch (1976). That is, a change is observed in the previously noted characteristic of income loss for those with higher education in Brazil (Ribeiro, Souza, Carvalhaes, 2010). In fact, a veritable race for more credentials is established, aiming to maintain or enter the best jobs.

Data and Methodology

To achieve these objectives, a quantitative methodology was used, based on the STATA statistical software. This process went through three distinct stages, which led to some answers to the main question of this study: a) database construction; b) use of descriptive statistics; c) application of inferential statistics. In the first stage, called database construction, an analysis and aggregation of different databases was carried out, with reference to the supplement on social mobility of the National Household Sample Survey (PNAD). The application of this survey supplement aims to question respondents about the employment and education level of their parents. Thus, this study was carried out by aggregating the supplement on social mobility from the 1982, 1996 and 2014 surveys. The use of this criterion is based on the relationship between the availability of cohort control and the availability of the surveys carried out.

By aggregating these cross-sectional studies into a single database, the occupational variables of the respondents and their parents were reordered based on the EGP occupational typology (Erikson-Goldthorpe-Portocarero, 1979). This typology comprises a series of arrangements for defining the categories used; however, a division into six categories was adopted [Professionals and Administrators, Non-Manual Routine Workers]. Also in the construction of the database, for the application of the binomial logistic regression statistical technique, a middle-class variable was created, considering two different possibilities: a) use of the occupational category "Professionals and Administrators"; and b) use of the category "Professionals and Administrators". Administrators + Non-Manual Routine Workers. Despite numerous discussions about the categorization of the middle class in Brazil (Ribeiro, 2012), based on the use of the EGP

typology, we opted to follow the work carried out by Salata and Scalón (2012), in order to encompass a greater differentiation between individuals in this occupational category in relation to income and education. In addition to selecting the occupational variables of the respondents and their parents, based on the EGP typology, we considered the following selection criteria for the data sample for intergenerational comparison purposes: 1) family situation as head of household; 2) age range between 30 and 45 years, considering the possibility of completing higher education and entering the job market (Breen, Muller, 2020).

Furthermore, for the construction of educational variables, the CASMIN method of completed years in the educational system and the educational transitions/credentials method were used. For the educational transitions/credentials variable, the construction already carried out by Ribeiro (2023) in another study was used. Thus, the variable comprises 4 categories: elementary school I (0 to 4 years), elementary school II (5 to 8 years), high school (9 to 11 years) and higher education (12 years or more). For classification by level of schooling, criteria were used that relate the position of the most schooled individual to the least schooled, which were separated into four categories corresponding to the relative frequencies of each year: low schooling, medium-low schooling, medium-high schooling and high schooling. The total sample consists of 10,544 cases, distributed across the four years of the research already mentioned.

The second part of the methodology involves the application of descriptive statistics of two orders: 1) cross-tabulation of the years of study of the population, either in absolute terms or disaggregated by type of occupation; 2) use of intergenerational mobility tables with analysis of migratory flows. The objective is to observe, in general, how data on educational credentials are established over time, as well as to describe the mobility that has occurred in recent decades.

Finally, in the third part of the methodology, the inferential technique called Logistic Regression (Xie; Powers, 2000) is applied. The use of this technique aims to analyze the chances of upward intergenerational social mobility or the permanence of highly educated individuals in the middle class. Based on this analysis, we relate educational credentials – in absolute terms – in order to understand the mechanisms that relate these variables, observing how educational credentials have increased or decreased the chances of social mobility. Seven different models were tested in order to observe the best fit of the data with the model and the sociological interpretation of the phenomenon of educational expansion. In addition to the occupational category of Professionals and

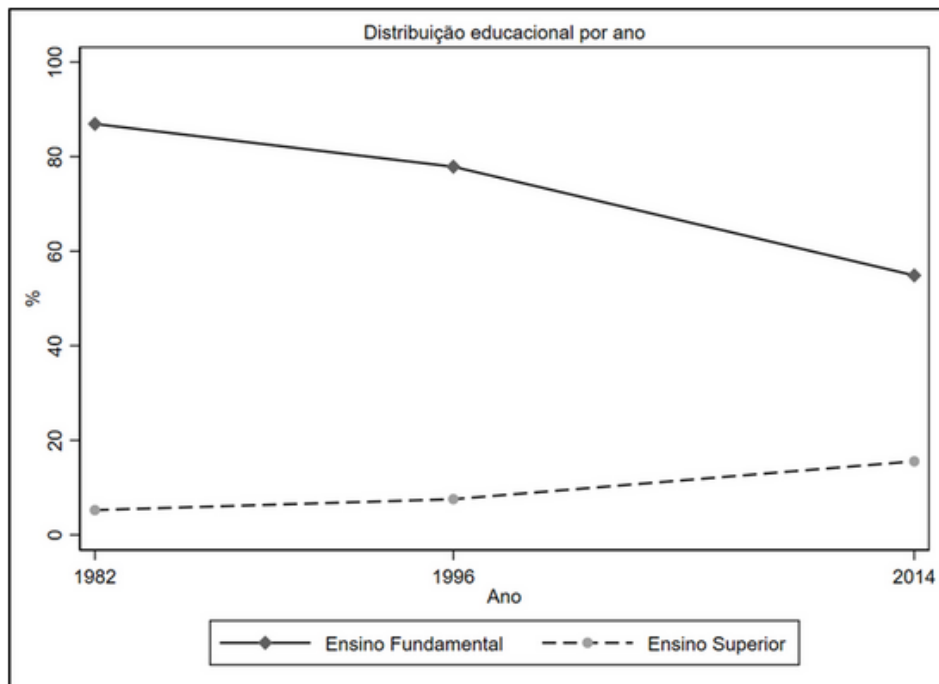
Administrators, used to observe the chances of remaining in the middle class, we analyzed in more detail the chances of the category of Unskilled Manual Workers ascending to the middle class, as ideal types in the formation of the class structure in Brazil. For post-estimation analysis of the data, average marginal effects (EMM) were used.

Analysis and Results

As mentioned in the previous section, the analyses are based on three steps: a) descriptive analysis of the educational distribution in the Brazilian population; b) mobility tables based on entry and exit flows; c) analysis of the chances of entering and remaining in the middle class.

In the figure below, the population distribution by educational level was estimated among the surveys applied (1982, 1996, and 2014). This graph indicates the extent of the educational expansion that has occurred in Brazil, especially in recent decades. The comparison aims to highlight how the educational level of Brazilians has grown over time, with a decrease of more than 20% in the proportion of people in primary education, as well as an increase of almost 10% in the population that is in higher education (incomplete or complete).

Figure 1 – Educational Distribution by Survey



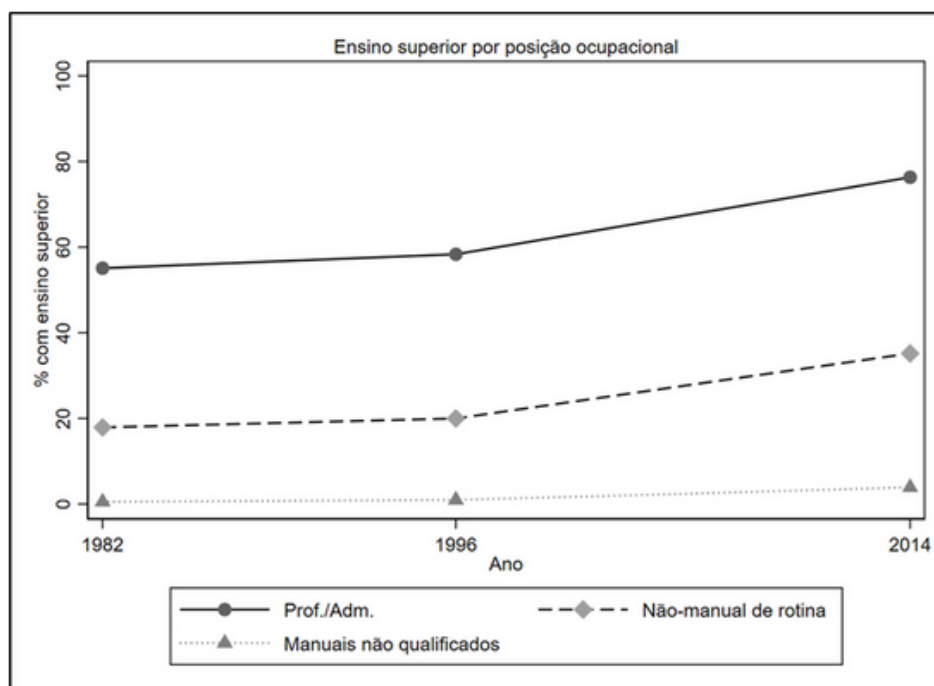
Source: Produced by the author, based on microdata PNAD/IBGE

This data shows not only the growth in the global educational level, but also highlights how progression within the educational system has become more fluid. This is a fundamental characteristic of the phenomenon of educational expansion around the world, and it has been verified, albeit belatedly, in the Brazilian case.

In view of this global growth in the educational level, it is now necessary to analyze how this trend is verified from the occupational position of individuals. Based on the greater interest in the higher education levels of individuals, as shown in the graph below, we seek to understand how the relationship between population distribution and Higher Education is established by the occupational position of the respondents (social destiny).

Thus, it was found that the trend of increasing the proportion of individuals with higher education in the highest occupational positions was more pronounced than in the positions that make up the base of the stratification pyramid in Brazil. In the categories of Professionals and Administrators and Non-Manual Routine Workers, there was a 20% increase in the proportion of each category, while the category at the base of social stratification, Unskilled Manual Workers, had a lower increase.

Figure 2 – Population distribution with higher education by occupational position

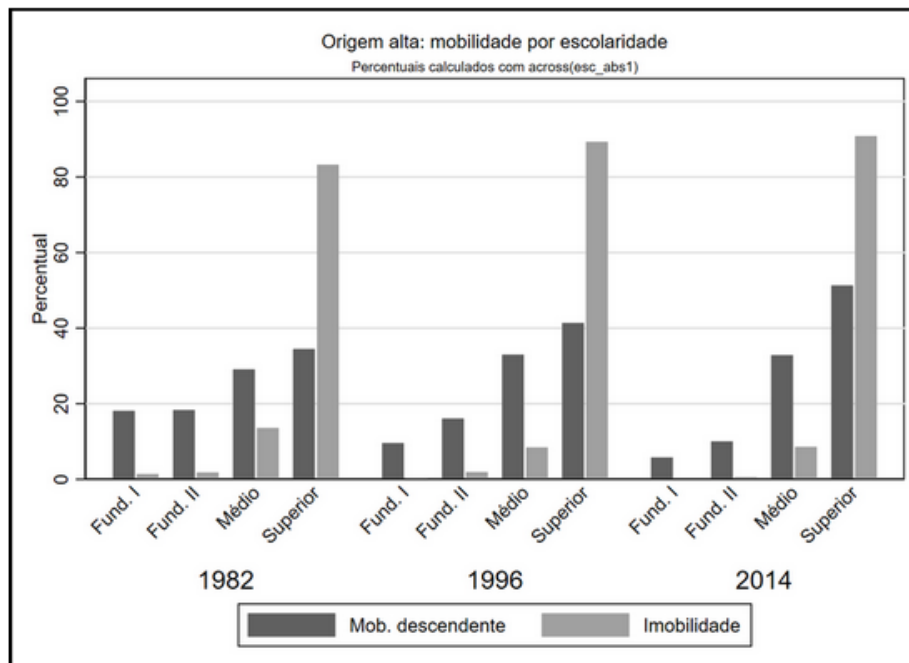


Source: Produced by the author, based on microdata PNAD/IBGE

This result already seems to indicate a trend regarding the relationship between social destiny and educational level, an argument already established in the literature of the sociology of stratification. However, how is this relationship established with reference to social origin and schooling?

In the graph below, a population distribution by education level is carried out, based on the social origin of the upper middle class (Professionals and Administrators). Considering the relationship between origins and destinies, we begin to analyze the social destiny of the upper middle class, considering schooling. As can be seen in the graph, we observe two trends in this process, the first around the high immobility of those with Higher Education that is maintained over time, that is, they managed to keep their children in the highest strata of society. At the same time, the proportion of individuals who had downward mobility increased for those with Higher Education. This trend reveals how the educational level becomes relevant in the process of social stratification, but less and less a guarantee for remaining in the highest strata.

**Figure 3 – Population distribution by education level, based on social origin:
Professionals and Administrators**

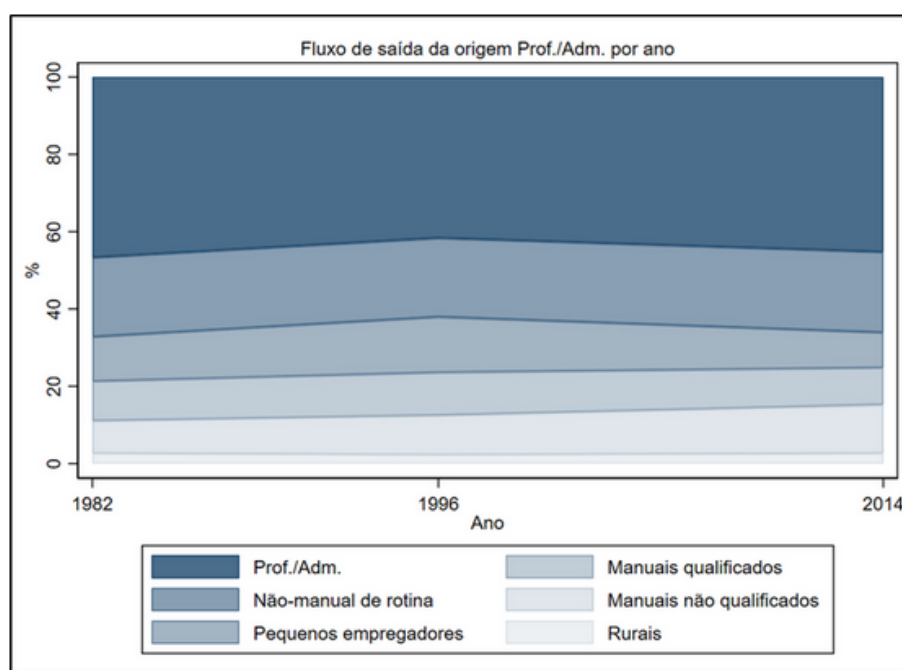


Source: Produced by the author, based on microdata PNAD/IBGE

In this sense, after analyzing the descriptive statistics concerning the proportion of individuals with a higher education level, we moved on to the second stage of the analysis. In this part, we sought to understand how the flows of entry and exit from the middle class were established (here considering the groups concerning the EGP – Professionals and Administrators).

As shown in the graph below, when we analyze the entry and exit of occupational groups, we begin to see another important aspect of social stratification. On the one hand, there is a certain homogeneity in the fate of individuals who originated in the middle class over time. This trend indicates, even if still descriptively, a strong relationship between social origin and destinies. That is, on the one hand, we see a substantial increase in the educational level of the population, but, considering individuals who came from the middle class, we can see that there is a maintenance in social destinies.

Figure 4 – Outflow from the occupational source Prof. and Adm. (middle class)

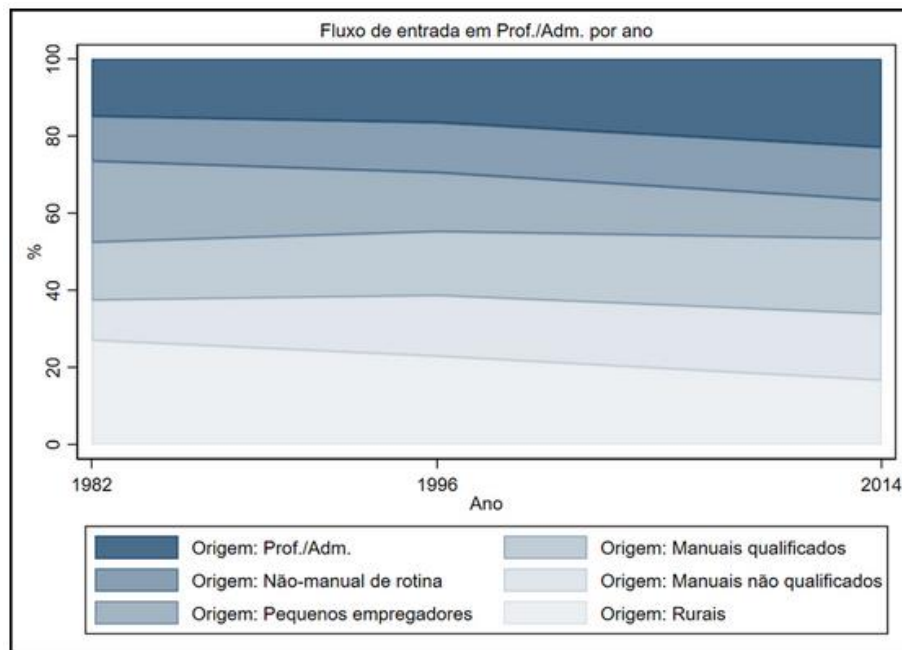


Source: Produced by the author, based on microdata PNAD/IBGE

On the other hand, when we consider the inflow, that is, the main social origins that make up the social destiny of the middle class, we can observe an increase in the percentage of individuals from the middle class itself and non-manual routine workers. In the strata at the base of the pyramid, there is a process of displacement from rural to urban occupations, with a decrease in the category of Rural Workers and an increase in

Unskilled Manual Workers and Skilled Manual Workers. This change is what is considered in the literature to be short-range displacements occurring along social stratification, especially due to the decrease in Rural Workers in the composition of the strata.

Figure 5 – Inflow to the occupational destination Prof. and Adm. (middle class)



Source: Produced by the author, based on microdata PNAD/IBGE

These trends indicate a process of reconfiguration of inequalities, especially in the process of forming the middle class. To some extent, we analyzed a path of maintaining inequalities based on social origin, as well as their reconfiguration. This reconfiguration lies in the decrease in social recruitment from the lower strata, as observed in previous years, to the detriment of class self-reproduction observed in the middle class.

From the descriptive analyses and mobility tables, we were able to observe two trends: a) an increase in the educational level of the population; b) the maintenance of social origin as a conditional factor for social destinies. These processes help us to compose the context regarding the analysis of the chances of entering and remaining in the middle class.

In view of the latter point, regarding the chances of entering and remaining in the middle class, 7 different logit models were used to understand which result best fits the data provided. In addition, post-estimation techniques were performed to assess the

chances of entering and remaining in the middle class. These results were estimated for individuals from three different configurations: 1) mixed; 2) men; 3) women.

Table 1 – Relationship between tested Logit models

Model	Descript	N	Df	AIC	BIC
M1	Origins	10.544	2	5200815	5200830
M2	M1+ Survey	10.544	4	5199274	5199303
M3	M2+ Education	10.544	7	4156421	4156472
M4	M3+ control	10.544	14	4135253	4135354
M5	M4+ Interactive Effect between Origins and Survey	10.544	16	4125398	4125514
M6	M4+ Interactive Effect between Origins and Education	10.544	17	4133006	4133130
M7	Saturated	10.530	30	4100698	4100916

Source: Produced by the author.

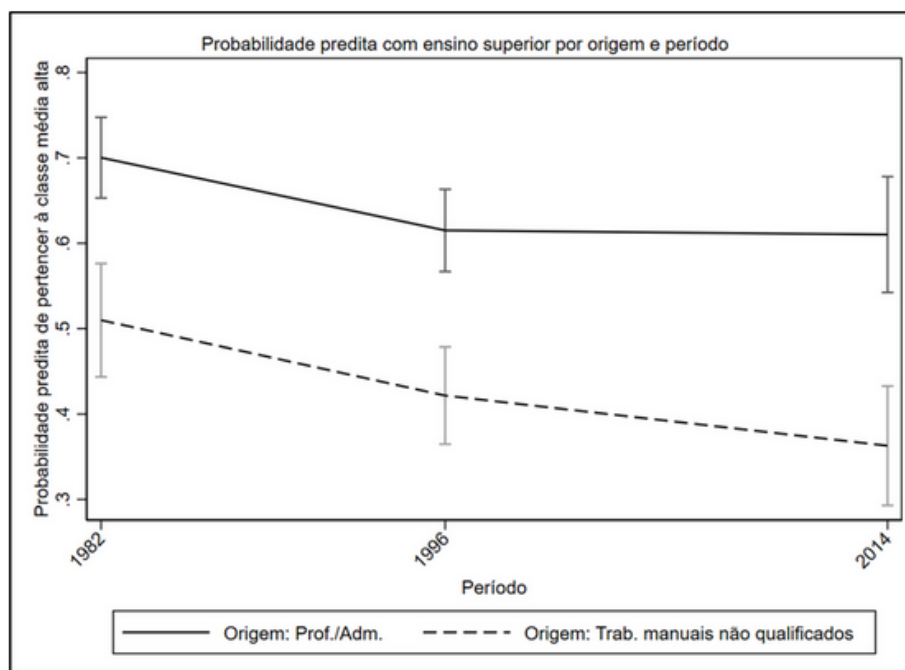
Analyzing the models, we can observe that the saturated model shows a better fit with the variables used, despite the drop in N. In other words, the change in chances would involve the relationship between origins, education, and the time of application of the survey. Disregarding the saturated model, we can consider that the model that best fit is M5, to the detriment of M6. This result indicates that educational expansion, analyzed in isolation, contributes very little to a change in the logic of opportunities. Instead of this mechanism being established, what was more preponderant for the observed social change is the relationship between social origins and the occupational structure observed over time.

Although the model fit indicates this composition as the most preponderant, we consider analyzing some post-estimation techniques to understand the relationship between education and social mobility. Here at this point in the research, we use two ideal types: 1) Professionals and Administrators as middle class; 2) Unskilled Manual Workers as the base of social stratification in Brazil. These techniques were used in order to understand the chances of entering and remaining in the middle class, considering individuals who are at the highest educational levels, in this case, Higher Education. Considering these prerogatives, we used the predicted probabilities technique for the following groups of individuals: a) mixed; b) women; c) men.

In this way, considering individuals of both sexes, in the table below, we can see that the chances of individuals in the middle class remaining in this stratum have tended to fall, but have remained stable in recent times. As for the base of social stratification,

individuals with social origins as Unskilled Manual Workers, the downward trend in the chances of entering the middle class continues.

Figure 6 – Predicted probabilities of individuals with higher education entering the upper middle class by social origin (mixed)

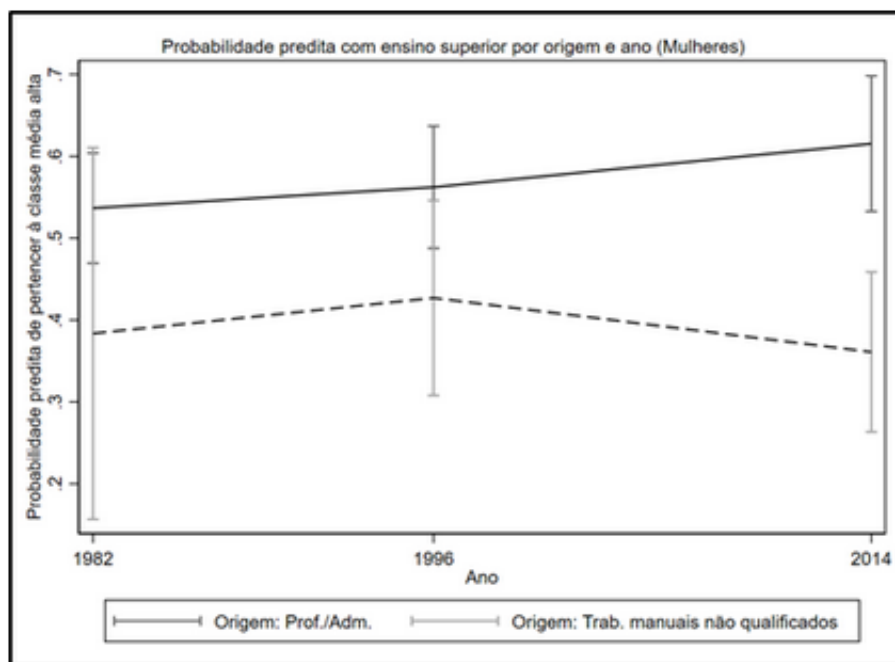


Source: Produced by the author, based on microdata PNAD/IBGE

After separating the analyses based on the individuals' sex, we can observe a significant change. Despite the greater complexity in interpreting the data, especially due to women's late entry into the labor market, we can observe a trend of greater chance of remaining in the middle class for sectors originating from the Professionals and Administrators class. In addition, there is a downward trend in the chances of women who came from the base of the stratification entering the middle class.

This late entry of women into the labor market, especially in the top sectors of the pyramid, indicates that there is still a remnant of structural change for this social group. Whereas for sectors of social origin of Unskilled Manual Workers, this expansion trend has stabilized. It should be noted that the interpretation of such data is more difficult, especially considering the margin of error of each parameter observed.

Figure 7 – Predicted probabilities of individuals with higher education entering the upper middle class by social origin (women)

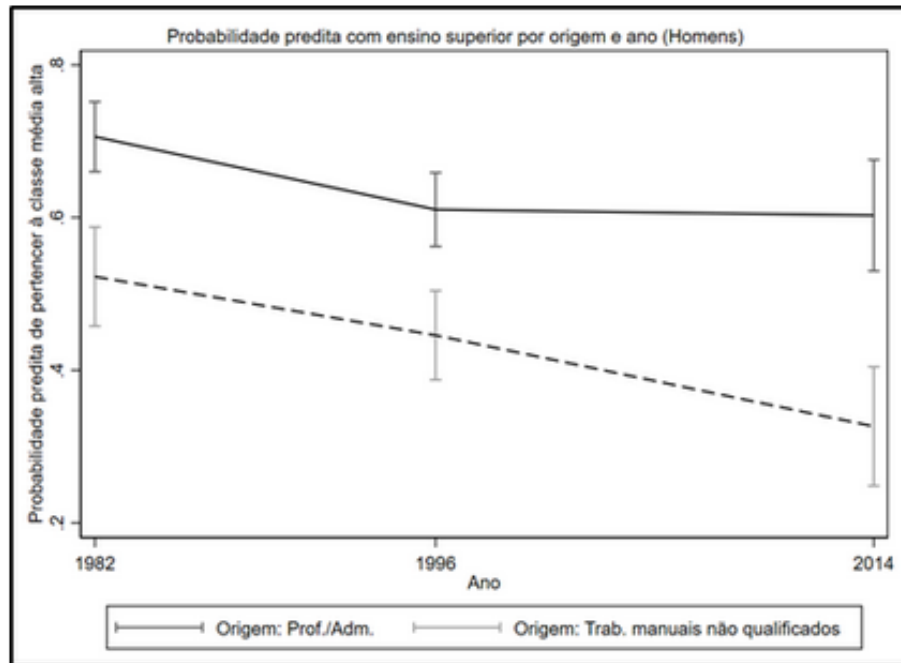


Source: Produced by the author, based on microdata PNAD/IBGE

When considering only the social group of men in the analysis of the chances of entering and remaining in the middle class, we observe a more consolidated trend of a downward trend in the chances of middle-class members remaining in this social stratum. Here we highlight the trend, taking into account the established margin of error. However, when we consider sectors originating from the social background of Unskilled Manual Workers, we observe a drop of about 15% in the chances of entering the middle class.

These factors indicate that completing high school and entering higher education does not guarantee, as in other times, entry into the highest sectors of society. This does not mean that education does not matter for social mobility, however, the factors linked to educational expansion, implemented in isolation, do not help to reduce social inequalities. More than ever, inequalities of opportunity must be based on inequalities of conditions to obtain more promising results.

Figure 8 – Predicted probabilities of individuals with higher education entering the upper middle class by social origin (men)



Source: Produced by the author, based on microdata PNAD/IBGE

Conclusions

Considering the results obtained by this study, we can observe four factors:

- Continuous process of increasing the value of credentials in the job market;
- Due to the universalization of education at the basic level (especially primary education), there is a process of shifting the value of educational credentials to higher levels (secondary and higher education);
- The process of increasing the value of educational credentials in the job market to overcome the barriers of social origins coexists with the tendency for a relative loss of the value of educational credentials in the process of social mobility.
- The relative decrease in the chances of mobility via educational credentials can lead to a series of factors: a) deepening and crystallization of inequalities based on the perception of educational credentials as merit in society; b) reconfiguration of individuals' perception of the advantages of pursuing higher education, centralizing options for the imperial professions (engineering, law, and medicine).

These last two analyses will be explored in the second phase of my doctoral studies, in order to understand how inequalities of opportunity are established and perceived in Brazil, especially the relationship between education and social mobility.

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